

POST: Pruning Oriented Security for Inversion Attack in Edge-Based Internet of Things

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Abstract—With the recent emergence of artificial intelligence (AI), edge users in industries and manufacturing have been extensively using AI-based services, which pose privacy and security risks to data. Distributed learning approaches deployed in the manufacturing industries help reduce data risks. However, traditional approaches cannot handle deep models and the scalability of edge-based Internet of Things (E-IoT) devices, especially in the manufacturing sector. Studies have proposed SplitFed learning (SFL) by combining split learning (SL) and the federated learning (FL) paradigm, but they fail to achieve an optimal tradeoff between communication and computational limitations and are vulnerable to inversion attacks. We present a pruning-oriented security (POST) method that is designed around the SFL paradigm that not only helps in achieving a balance between communication and computation load for E-IoT devices but also preserves the data and model privacy against inversion attacks. The POST leverages the concept of using a higher number of layers in the E-IoT devices, which restrains the attacker from reconstructing the outputs. Furthermore, the POST adds communication and computation constraints in the optimization function to reduce the overall cost of the method. The novel pruning method adopts the regularization and adversarial training approach to further improve the preservation of the privacy of intermediate features in the SFL paradigm. We conduct our experiments on publicly available datasets in real-world settings to illustrate the efficacy of the POST method in terms of preserving privacy while ensuring the best tradeoff for communication and computation load.

Index Terms—Data privacy, edge-based Internet of Things (E-IoT), inversion attacks, split learning (SL), split federated learning.

I. INTRODUCTION

THE development of Internet of Things (IoT) devices and their deployment in industries and the manufacturing sector have paved the way for researchers and companies alike to acquire large amounts of data in a ubiquitous manner. Data from IoT devices, such as sensors, wearables, and smartphones, used in the manufacturing sector are used to train artificial intelligence (AI) models to provide automated services [1]. The acquired data can also be used to fine-tune existing trained models to boost the performance of the AI trained model. With the rapid progression in the field of semiconductor devices and processors, edge devices are now capable of deploying storage and computing resources, thanks to the multiaccess edge computing (MEC) framework [2]. The edge deployment can help in reducing the latency, enabling real-time processing of applications, and providing high-bandwidth service to the stakeholders associated with manufacturing sector. Recently, distributed MEC systems have been proposed [3] that bring the intelligence to the edge components allowing them to train AI models locally or storing notable data volume collected through IoT devices for analysis at cloud server. In the manufacturing sector context, the edge devices might refer to smartphones, workstations, and IoT platforms that connect multiple sensors such as Raspberry pi, whereas the server refers to the central server or cloud server. Such systems in this article are referred to as Edge-based Internet of Things (E-IoT) and cloud server, respectively. Although many benefits have been recorded for using E-IoT, especially for AI-based services, E-IoT faces three main challenges. The first is related to communication and resource constraints, the second is related to underperforming AI models, while the third is related to data security. For the first issue, the collected data from IoT devices are heterogeneous in nature, and therefore, the conventional centralized learning paradigm only uses edge devices to store the data and send it to the cloud server. The cloud server is then responsible to perform in-depth analysis and recommendation-based services [3], [4]. The transmission of data to the cloud requires constant communication and bandwidth while adding incremental latency. Such constraints lead to performance degradation in terms of accuracy and timely services. The second issue relates to the local training of the model at edge device as the model can only be trained on a limited number of samples due

Received 12 May 2025; revised 20 July 2025, 25 August 2025, and 2 November 2025; accepted 10 November 2025. Date of publication 13 November 2025; date of current version 26 January 2026. This work was supported in part by the Deanship of Graduate Studies and Scientific Research at the University of Bisha through the Fast-Track Research Support Program, in part by the European Union's (EU's) Horizon 2020 the Marie-Curie under Grant 847652, and in part by the Science Foundation Ireland under Grant 12/RC/2276_P2. (Corresponding author: Abi Waqas.)

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Digital Object Identifier 10.1109/JIOT.2025.3632564

to the computational and storage constraints. Such models introduce unstable training and performance degradation due to insufficient data and computational constraint. The third challenge corresponds to the user privacy as the data are sent to the server for training to stabilize the training process and improve the performance of the service.

As the uploaded data can be private in nature, it poses potential privacy and security risks concerning users' data. In recent times, institutions and users have shown unwillingness toward their private data sharing to the Internet companies and services providers due to the risk of data breaches [5]. With the challenges mentioned above, the federated learning (FL) paradigm was introduced which handles data privacy concerns to some extent.

FL [6] paradigm has garnered a lot of attention from research community due to its privacy preserving characteristics as it allows multiple clients (users) to train the network locally without sharing the data to the server. The locally trained models from multiple users are then sent to the server for global aggregation. The updated model from the global aggregation is then sent back to each user, accordingly. The problems with FL paradigm are impracticality of training and inference on edge devices in E-IoT environment due to the storage and computational constraints [7]. Furthermore, FL addresses data privacy issues but as all stakeholders involved in the training process have access to entire model, it is vulnerable to model privacy issues such as membership inference, model poisoning, and model inversion attacks.

To address the limitation of FL paradigm, split learning (SL) method was introduced [8] that works in a distributed manner similar to FL. However, SL offloads a part of the training process to the server through model division in server side and client side, hence the name SL. The server-side and client-side models pass the intermediate features to perform backpropagation and forward computation without retrieving each other's models. Although it solves the practical issues of training process on the edge devices, it is still vulnerable to attacks that are related to model and intermediate features such as feature space hijacking attack, gradient leakage attack, and the ones mentioned above. Furthermore, SL is slower than FL as the training process is performed sequentially, which introduces significant delays in the context of large-scale E-IoT devices.

Recently, an intent to combine the characteristics of split and FL paradigm was proposed in [9] that addresses the drawbacks of each of the paradigms in terms of computational complexity and latency. Unlike the FL, which requires full-model training on resource-constrained E-IoT devices (often impractical [9]), SplitFed learning (SFL) offloads 70%–80% of layers to the server. However, the SFL, like its predecessors, is also vulnerable to feature-related and model-related attacks, respectively. This work specifically focuses on addressing the inversion attacks in the SFL scenario.

The inversion attacks aim to reconstruct the private information from intermediate feature maps during the inference phase [10]. Inversion attacks are possible due to various concerns that include untrusted cloud server, unknown communication environment, or an attacker that can access the model through communication interception. The inversion attack raises seri-

ous concerns when it comes to model privacy. Existing studies have proposed two fundamental mainstream methods to cope with the aforementioned attacks. The first stream proposes cryptographic techniques such as homomorphic encryption [12] of secure multiparty computation [13], while the second stream proposes randomly dropping intermediate feature, adding perturbations for adversarial training, and injecting noise intentionally at the input side [14], [15]. From the two streams, the first one provides a better defense against inversion attacks, however, they also add significant delays to the end services. Research showed that the computational cost and delay can extend the slowdown by more than 10 000 times [16]. Such slowdown is impractical, especially for the E-IoT devices.

To address the limitations of existing defense methods in SFL paradigm, we propose pruning-oriented security (POST) method to provide a tradeoff between utility, efficiency, and privacy to ensure practicality for E-IoT devices. The reason for choosing SFL for the proposed POST is that it allows the POST to deploy deeper client-side models (enhancing privacy) while retaining edge feasibility. The POST operates on two assumptions to cope with the inversion attacks. The first assumption is that the initial layers are better at extracting users' information while the latter layers are good at extracting task-specific information, and therefore, adding more layers will create difficulties for the attacker to reconstruct the data effectively through the inversion attacks. To this end, the POST methods will add more layers at the client side before sending the intermediate feature maps to the server side in SFL paradigm.

The second assumption caters to two problems, which is the computation and storage constraints for E-IoT devices. Existing studies have shown that the network pruning for deep architectures has been performed extensively to improve the inference speed while maintaining a suitable performance requirement [17]. Existing research only focuses on the tradeoff between efficiency and accuracy when pruning the models. However, in this work, we want to focus on model pruning customized to maintain the privacy of the model. In this regard, POST introduces adversarial reconstruction training and regularization to improve the training stability while pruning the model. The model pruning will help at three fronts. The first is the real-world practicality of this method as E-IoT devices can handle pruned model. The second is the adversarial training using regularization with pruning as the regularization method is designed to increase the reconstruction error, and thus, the attacker will have a difficult time to reconstruct the data using inversion attacks. We assume that if the attacker is somehow able to participate the poisoned data, it will use the same architecture for the participation. Therefore, the third front is the identification of poisoned attack, as the attacker will participate with unpruned version of the model, and therefore, the server side will recognize the attacker model and will not use it for the global aggregation. The contributions of this work are summarized as follows.

- 1) We propose POST that enables deeper model deployment on resource-constrained E-IoT devices through

structured pruning, enhancing privacy against inversion attacks.

- 2) We proposed the combination of adversarial-regularization strategy that stabilizes pruning and amplifies reconstruction errors, directly addressing inversion vulnerabilities.
- 3) We show that the POST optimizes communication, computation, and privacy simultaneously, validated via experiments on publicly available datasets.
- 4) We achieve state-of-the-art results, i.e., superior compression rates ($\sim 66\%$ parameter reduction) coupled with reduced attack success rates.

II. RELATED WORKS

In this section, we consolidate the literature on distributed learning paradigm followed by the brief review of studies handling privacy and security issues with respect to inversion attacks.

A. Distributed Learning Paradigms for E-IoT

The FL paradigm has garnered a lot of attention since its release as it only shares gradients or model parameters instead of raw data, which maintains the users' data privacy to some extent. One of the key elements of FL paradigm is the global aggregation of the model gradients sent by the client. The study [6] proposed the federated averaging (FedAvg), one of the earliest aggregation methods for FL paradigm. The study in [18] proposed a reparametrization and generalization to the FedAvg method, i.e., FedProx for improvement. The study achieves this by adding a proximal term in the local optimization object at the client side and computing the L2 distance between the global and local model at each iteration. The distance impact on the proximal term constraints the client-side model to deviate much from the global model.

As mentioned earlier, FL paradigm has limitations in terms of privacy risks, communication costs, and computational capabilities concerning E-IoT devices.

The SL paradigm was introduced to address limitations of FL paradigm in [8]. In the SL paradigm, a cut layer is designed to split the network between the client and the server. The client feeds the data to the part of the model deployed locally, while the server gets the intermediate feature maps to perform task-oriented training. SL is preferred for E-IoT devices due to the enhanced model privacy and less usage of computational resources. Studies such as [19] and [20] tried to improve the privacy of SL paradigm using shadow models or combining it with graph neural networks. One of the limitations of SL paradigm is the scalability due to its sequential approach, for which studies use parallel training approach in a multiclient environment. However, SL methods are equally vulnerable to inversion attacks along with catastrophic forgetting [21] which is caused by forgetting previously learned information.

Recently, a study [9] proposed a hybrid approach named SFL to address the limitations of SL and FL paradigm. The SFL approach leverages the strengths of both the paradigms, i.e., the practicality of using E-IoT devices from SL and the parallel training approach from FL. The SFL paradigm also solves the catastrophic forgetting problem as all the clients

participate in the training process and contribute to global aggregation. The study proposed two versions of SFL which only differs in how the model is aggregated at the server side. A study in [23] compared the performance of SL, FL, and SFL paradigm under different settings. The study concluded that with independent and identically distributed (i.i.d.) data, SL performs the best; however, with non-i.i.d. data, SL performs the worst among all the aforementioned paradigms. Following this, studies have tried to improve specific aspects of the SFL paradigm. For instance, the study in [22] proposes SplitAVG method for the aggregation to handle heterogeneity, while the study in [24] proposes the PPSFL approach for addressing privacy and security. This work mainly focuses on the inversion attacks for SFL paradigm.

B. Inversion Attacks

Recently, inversion attacks are being studied extensively due to their ability to reconstruct private information from either the model gradients or intermediate feature maps [11]. Inversion attacks were originally proposed to gain insights from the training data while recovering some information; however, recent works illustrate that the inversion attacks pose a threat to privacy and security of trained AI models and user. The study in [11] proposed an inversion attack that could reconstruct the data from model output. The inversion attack first initializes an inversion network and then trains it in accordance with the target model output so that the initialized weights in inversion network are brought closer to the target model's weight. Similarly, the study in [14] proposed the inversion attack method for reconstruction of data from intermediate feature maps in an E-IoT environment. Researchers have also worked on generative network architectures such as GANs and autoencoders to perform the inversion attacks as proposed in [5] and [11].

Holistically, the defense methods against inversion attacks can be categorized into cryptographic-based methods and perturbation-based methods. Cryptographic methods include homomorphic encryption, trainable parameter encryption, and secure-multiparty computation (SMC) [25], while the perturbation-based methods include addition of noise, intentional adversarial attacks, or adversarial training strategies [14], [15]. Among both the categories, cryptographic methods provide a better defense against inversion attacks; however, such methods are impractical to apply on E-IoT devices due to resource constraints. Therefore, for E-IoT devices, perturbation-based methods are widely used as well as feasible for practical usage.

The study in [16] proposed the selection of features that least contribute to the inference results and addition of noise to the selected features to defend against inversion attacks. The study [14] proposed dropping skip connections and parts of output to defend against inversion attacks. However, the aforementioned methods severely degrade the prediction performance. Several studies have proposed adversarial training strategies to cope with the inversion attacks [26], [27], [28]. The adversarial training strategy increases the difficulty of attackers when reconstructing the data via inversion attacks by enabling the model to show less information at the output. However, similar problems are faced by adversarial training

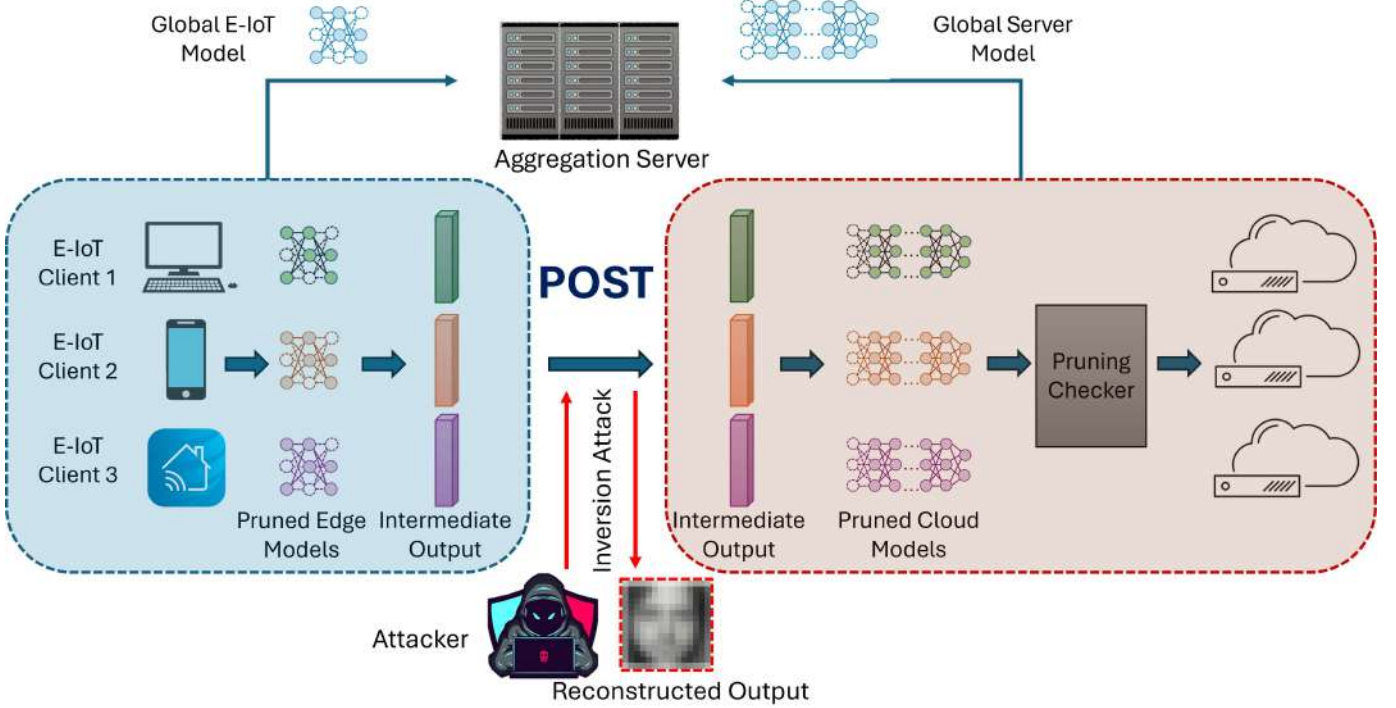


Fig. 1. Proposed POST framework for defense against inversion attacks in SFL paradigm.

strategies, i.e., reduced performance, due to the addition of larger perturbations. Therefore, in this article, we intend to propose an efficient pruning-oriented approach that is compliant with E-IoT resource constraints and maintains effective prediction performance.

C. Network Pruning

The aim of network pruning is to compress a network architecture through the removal of redundant or selective parameters for reducing the computational cost and improving the inference efficiency [28]. The pruning process is very intriguing for researchers working with E-IoT as it allows deep models to be used at edge devices with a practical inference speed. The study in [17] improved the model efficiency by removing the parameters yielding large magnitudes. Recently, soft-pruning methods have been introduced that make the use of mask to prune the model parameters. The study in [29] uses binary masks to prune the weights that are masked as 0 s while training. Yang et al. [30] proposed to add masks for each filter or neuron so that the parameters can be pruned in accordance with their importance. Lin et al. [31] focused on the channels, branches, and blocks instead of individual filters and neurons. The study proposed to design a soft mask that analyzes the structures, identifies importance of each structure, and prunes the less important ones. This study is inspired by the work proposed in [28] and [31] to design the POST method for the defense against inversion attack in SFL paradigm.

III. METHODOLOGY

The overall workflow of the proposed POST for SFL paradigm is shown in Fig. 1. In SFL paradigm, each E-IoT

client will train the model locally for their network split and send the intermediate output to the server side. The server will then continue the training using their network split with the intermediate feature maps received from the client side. The aggregation server collects the data from both the E-IoT clients and the server and constructs a global model for each of the entities. We propose POST to cope with the inversion attack using the pruned edge and pruned cloud models along with a pruning checker that would determine whether the data are pruned exactly as it is supposed to be. In Sections III-A–III-D, we provide details regarding the preliminary details, threat model, and design method, accordingly.

A. Preliminary Details

The SFL paradigm comprises E-IoT clients and a cloud server. Let us assume m clients. Then, the local data with the E-IoT clients and associated label can be represented as (X_m, y_m) . The SFL paradigm uses the collaboration of $m \in M$ clients with their resources and local data to train and optimize the model θ jointly for the global loss $F(\theta)$ minimization. The formulation for the same is given in the following equation:

$$\argmin_{\theta} F(\theta) = \argmin_{\theta} \frac{\sum_{m=1}^M F_m(\theta)}{M} \quad (1)$$

where the local loss $F_m(\theta)$ for each client to minimize the global objective is defined in the following equation:

$$F_m(\theta) = \frac{\sum_{x \in X_m} \text{loss}(\theta; x)}{d_m} \quad (2)$$

In (2), each E-IoT client's data sample is denoted by $d_m | d_m = |X_m|$ and the prediction loss on the training sample is denoted by $\text{loss}(\theta; x)$.

B. Threat Model

The representation of the attack is shown in Fig. 1, where an adversary exploits intermediate feature maps transmitted to the cloud server for reconstructing the raw inputs (e.g., surveillance images). We assume a closed-box inversion attack wherein the attacker has zero access to E-IoT model parameters or cloud server internals. We assume that the attacker can intercept intermediate features during client-server communication. We assume that the attacker possesses prior knowledge of training data distribution, E-IoT/client model architectures, input dimensions, and cloud topology, which is in compliance with the existing studies [9], [11], [20], [24]. We also assume that the attacker cannot compromise E-IoT devices or the cloud server physically. The E-IoT devices constraints (limited storage/computation) prevent encrypting features or storing full models, forcing reliance on lightweight defenses. The proposed threat model is representative of practical E-IoT deployments where attackers exploit communication channels but lack system-level privileges.

C. Rationale for Pruning as a Defense

The proposed POST method uses pruning, which we assume to be a lightweight but an effective defense strategy against the model inversion attacks. We base our assumption on the fact (suggested by existing studies) [28], [31] that the pruning increases the sparsity of the feature maps, which, in turn, increases the entropy. Pruning does that by removing redundant structures and parameters from the networks, and thus, the less amount of information can be recovered through the transmitted features. Therefore, by the aforementioned assumption, pruning increases the difficulty in reconstructing the raw inputs. The sparsification from the pruning process disrupts the low-dimensional manifolds, which are mainly exploited by the inversion models to reconstruct the raw inputs. Although the cryptographic methods such as homomorphic encryption or secure multiparty computation, do work resisting model inversion attacks; however, they are computationally more complex than the pruning process, and therefore, they are not feasible for the E-IoT devices. On the other hand, differential privacy techniques, such as adding noise to the model utility also resists model inversion attacks but it also degrades the performance. Therefore, for the proposed method, we chose pruning method to be the primary focus of the POST method.

D. Design Method

The proposed method aims to cope with inversion attacks while maintaining the practicality of adding more layers at E-IoT devices without compromising the performance. This idea is based on the intuition that sensitive information is difficult to extract from the latter layers in comparison to the earlier layers as shown in Fig. 1. However, it is not practical to use more layers at E-IoT devices due to computational and storage constraints. Therefore, we adopt a structured pruning method [30] that can help add more layers at E-IoT devices within the aforementioned constraints. Studies have shown that the structured pruning helps accelerate the inference process

and sparse matrix computations by removing specific blocks or channels from the network architecture. Let us denote the deep neural network architecture with θ . As defined the model is split between the E-IoT devices and cloud server, and therefore, the E-IoT partition is denoted as θ_{et} , and the cloud server partition is denoted as θ_{cs} . We represent the parameters of each split model as ζ_{et} and ζ_{cs} , respectively. The overall model can be represented as $\theta = \theta_{et} \circ \theta_{cs}$. We propose a soft mask $\mathbf{SM}$, which will be trained to perform the pruning of network architecture such that the output of the structures are scaled. We apply the pruning to the block and to the channels. The pruning for block will consider the remapping of residual blocks while the pruning for channels will undertake the output of each channel. The $\mathbf{SM}$ will be trained in a manner such that the prediction performance is not affected by the pruning process. The proposed optimization function for training $\mathbf{SM}$ is shown in the following equation:

$$\min_{\mathbf{SM}} \text{loss}(\zeta, \mathbf{SM}) + \lambda_1 \|\mathbf{SM} \odot \alpha(\zeta)\|_1 + \lambda_2 \|\mathbf{SM} \odot \beta(\zeta)\|_2 + \lambda_3 C(\mathbf{SM}) + \lambda_4 E(\mathbf{SM}, \zeta). \quad (3)$$

The proposed optimization function in (3) combines multiple objectives to achieve a balance between model performance, sparsity, communication efficiency, and energy consumption. Each term was designed to address one of the aforementioned constraints. The first term $\text{loss}(\zeta, \mathbf{SM})$ is the base prediction loss, which determines the prediction performance. The second term $\lambda_1 \|\mathbf{SM} \odot \alpha(\zeta)\|_1$ is a sparsity regularizer that reduces the model size through promoting structural sparsity. The term $\alpha(\zeta)$ makes the regularization adaptive to parameter importance and is defined as $\alpha(\zeta) = |\nabla_{\zeta} \text{loss}(\zeta)| / \max(|\nabla_{\zeta} \text{loss}(\zeta)|)$. The numerator here computes the importance of each parameter with respect to model's performance and normalizes it with maximum gradient to maintain the scale consistency. The multiplication $\odot$ ensures that a stronger sparsification is performed for the parameters that are less important. The third term $\lambda_2 \|\mathbf{SM} \odot \beta(\zeta)\|_2$ uses L2 regularizer to control the magnitude of the soft mask while stabilizing the training process. We proposed the term $\beta(\zeta)$ for determining the structural awareness importance which is defined as $\beta(\zeta) = \exp(-\gamma * S(\zeta))$. The parameter γ is the temperature parameter while $S(\zeta)$ measures the structural importance such as connectivity in the network. The above term preserves important structural patterns in the network. The third term represents the communication cost function that penalizes patterns in the mask that leads to inefficient communication between the cloud server and E-IoT devices. The fourth term $C(\mathbf{SM})$ is defined as $C(\mathbf{SM}) = \|\Delta \mathbf{SM}\|_1$, where $\Delta \mathbf{SM}$ represents the change in mask values between the communication rounds. Finally, the fifth term $E(\mathbf{SM}, \zeta)$ corresponds to the energy efficiency, which accounts for the computational costs associated with E-IoT devices and is defined as $E(\mathbf{SM}, \zeta) = \sum (\mathbf{SM}_i * \text{cost}_i)$. The term cost_i corresponds to the computational cost for maintaining each parameter. To adaptively prune the model, we adopt the fast iterative shrinkage-thresholding [32] method to adaptively determine the threshold for the soft mask to prune the network.

1) *Regularization*: In addition to the above loss function, we add another term to increase the reconstruction error

while reducing the inversion attack stability. In this regard, we adopt the Lipschitz regularization [33] to deal with the inversion attack. The conventional Lipschitz regularization makes the function smooth such that the changes in output are proportional to that of the input. However, we are using Lipschitz for the anti-inversion goal, and therefore, with the regularization, we want to control the instability to make the reconstruction difficult while maintaining the models' performance. Therefore, for the function ρ , the formulation for the Lipschitz condition can be written as

$$g = \sup_{x_1 \neq x_2} \frac{\|\rho(x_1) - \rho(x_2)\|}{\|x_1 - x_2\|} \quad (4)$$

where the notation g corresponds to the Lipschitz constant. The term g can be derived using the following equation:

$$\|x_1 - x_2\| \geq \frac{\|\rho(x_1) - \rho(x_2)\|}{g}. \quad (5)$$

The derivation suggests that given a small difference in the output, the reconstruction error can yield large difference by increasing $1/g$. Therefore, the effect of inversion attack can be reduced by maximizing g . We adopt the method proposed in [28] to approximate the constant g . As suggested, we then modify the main optimization function by adding Lipschitz loss as a regularizer as follows:

$$\begin{aligned} \min_{\mathbf{SM}} \text{loss}(\zeta, \mathbf{SM}) + \lambda_1 \|\mathbf{SM} \odot \alpha(\zeta)\|_1 + \lambda_2 \|\mathbf{SM} \odot \beta(\zeta)\|_2 \\ + \lambda_3 C(\mathbf{SM}) + \lambda_4 E(\mathbf{SM}, \zeta) + \text{loss}_g(\zeta_{\text{et}}). \end{aligned} \quad (6)$$

2) *Adversarial Training*: To protect data privacy against the inversion attack concerning SFL paradigm, we need to mislead the attacker, and therefore, we will first create a surrogate model θ_{sg} with its associated parameters ζ_{sg} for the inversion task. This model is optimized such that the input data can be reconstructed from the intermediate feature maps $\theta_{\text{et}}(x, \zeta_{\text{et}})$. The parameters ζ_{sg} for the surrogate inversion model will be optimized to minimize the difference between the raw input sample x and the reconstructed data using adversarial loss loss_{sg} . However, we need to consider that the loss is defined in the context of SFL architecture. Therefore, each E-IoT devices will have access to the local intermediate feature map only. Let us denote the intermediate feature map with an E-IoT device as $\theta_{\text{et},m}(x, \zeta_{\text{et},m})$. The adversarial loss for the SFL paradigm for the reconstruction of the output from an E-IoT can be formulated as shown in the following equation:

$$\text{loss}_{\text{sg}}(\zeta_{\text{et},m}, \zeta_{\text{sg}}) = \|x - \theta_{\text{sg}}(\theta_{\text{et},m}(x, \zeta_{\text{et},m}), \zeta_{\text{sg}})\|_2. \quad (7)$$

The integration of the surrogate adversarial model will make sure that the attacker is misled without compromising the prediction performance. Therefore, the overall optimization function can be rewritten as shown in the following equation:

$$\begin{aligned} \min_{\mathbf{SM}} \text{loss}(\zeta, \mathbf{SM}) + \lambda_1 \|\mathbf{SM} \odot \alpha(\zeta)\|_1 + \lambda_2 \|\mathbf{SM} \odot \beta(\zeta)\|_2 \\ + \lambda_3 C(\mathbf{SM}) + \lambda_4 E(\mathbf{SM}, \zeta) + \text{loss}_g(\zeta_{\text{et}}) \\ - \max_{\zeta_{\text{sg}}} \delta \text{loss}_{\text{sg}}(\zeta_{\text{et},m}, \zeta_{\text{sg}}). \end{aligned} \quad (8)$$

It should be noted that we need to consider the computational constraints associated with the E-IoT devices and (7) requires

bilevel optimization, which is computationally complex, thus not suitable for the E-IoT devices. Therefore, we replace the ζ_{sg} with $\hat{\zeta}_{\text{sg}}$ suggesting that the precomputed surrogate model trained for adversarial attack will be periodically updated by the central server rather than optimizing the function on each E-IoT device. The overall training process for the proposed POST is shown in Algorithm 1.

Algorithm 1 POST Training in SFL Paradigm

Require: E-IoT clients M , cloud server, pruning ratio, hyper-parameters λ_1 - λ_4 , δ

Ensure: Pruned global model θ

- 1: Initialize global model $\theta = \theta_{\text{et}} \circ \theta_{\text{cs}}$
 - 2: **for** each communication round $t = 1$ to T **do**
 - 3: Randomly select subset of clients M_t
 - 4: **for** each client m in M_t in parallel **do**
 - 5: Receive global mask $\mathbf{SM}$ from server
 - 6: Perform local forward pass: $f_m = \theta_{\text{et}}(x_m)$
 - 7: Compute local loss and gradients
 - 8: Update local mask $\mathbf{SM}_m$ using (3)
 - 9: Send pruned features and mask to server
 - 10: **end for**
 - 11: Server aggregates masks and features
 - 12: Update global model and mask via (8)
 - 13: Periodically update surrogate inversion model
 - 14: **end for**
 - 15: **return** θ
-

IV. EXPERIMENTS AND ANALYSIS

In this section, we first define our experimental setup, data partitioning, implementation details, performance analysis in terms of defense, prediction performance, communication overhead and efficiency, and computational cost.

A. Experimental Setup

In this work, we consider two inference tasks for evaluation. The first one corresponds to the vehicle reidentification from surveillance cameras using multiple E-IoT devices and a cloud server. An image is acquired by the E-IoT device, which is then processed through the split part at E-IoT. The outputs from the split part, i.e., intermediate feature maps, are then sent to the server. The experiment is conducted on VERIWILD [34] dataset, which is one of the largest dataset for vehicle reidentification comprising of 40671 vehicle and 416314 images. The images are captured during one month from 174 cameras. We perform experiments on different sizes of the VERIWILD datasets, i.e., small, medium, and large. The second is the COVID-19 radiography dataset [35], which is used for COVID-19 diagnosis. This is an open access dataset comprising of 15153 images.

For both the datasets, we choose ResNet18 as our backbone model for the training and identification tasks. The reason for choosing ResNet18 is the resource constraint nature of edge devices as the whole model cannot be deployed on edge. From the deep learning family, ResNet so far is the most feasible option for the testing and deployment at the edge

TABLE I
PERFORMANCE ANALYSIS OF POST ON THE COVID-19 AND VERIWILD DATASETS IN TERMS OF SSIM DROP, PSNR DROP, PREDICTION PERFORMANCE DROP, AND ATTACK PERFORMANCE DROP. THE POST IS COMPARED WITH EXISTING APPROACHES TO VALIDATE ITS EFFICACY

Method	Dataset	SSIM drop	PSNR drop	Prediction Accuracy drop	Attack Accuracy drop
FL [6]	COVID-19	2.3%	2.4%	7.1%	1.7%
	VERIWILD-S	1.9%	1.8%	8.2%	1.5%
	VERIWILD-L	2.4%	2.5%	11.0%	1.8%
SL [8]	COVID-19	2.5%	2.6%	6.4%	2.4%
	VERIWILD-S	2.2%	2.3%	7.6%	2.3%
	VERIWILD-L	2.8%	2.9%	10.5%	2.7%
SFL [9]	COVID-19	2.3%	2.4%	6.9%	1.9%
	VERIWILD-S	2.1%	2.2%	7.9%	2.4%
	VERIWILD-L	2.5%	2.5%	10.8%	2.8%
SplitAvg [22]	COVID-19	2.2%	2.3%	6.8%	1.3%
	VERIWILD-S	2.0%	2.2%	7.7%	1.5%
	VERIWILD-L	2.3%	2.5%	10.6%	1.9%
FedProx [18]	COVID-19	3.2%	3.3%	7.3%	1.9%
	VERIWILD-S	2.9%	3.0%	8.4%	2.4%
	VERIWILD-L	2.7%	2.8%	11.3%	2.6%
POST	COVID-19	12.3%	12.7%	5.3%	28.9%
	VERIWILD-S	14.7%	15.0%	1.8%	21.3%
	VERIWILD-L	14.5%	14.9%	2.7%	31.4%

devices. The assumption is in compliance with existing studies [9], [24], [28]. ResNet18 was pruned to a computational budget of 0.8 GFLOPs (suitable for an edge device). The energy term (λ_4) in (3) was scaled to device power profiles (≤ 5 W), with costs derived from MLPerf Tiny Benchmarks [37]. This study develops an inversion model with closed-box characteristics, which is designed to perform the inversion of network layers. For instance, if the E-IoT model has five 7×7 convolutional layers, then the inversion model will have five 7×7 deconvolutional layers. We also implement a surrogate inversion model, which takes the input from the last convolutional layer's output of E-IoT model. The inversion model is trained using the proposed adversarial regularization technique.

B. Data Partitioning

We partition datasets using a Dirichlet distribution for non-i.i.d. data splits that ensures the required level of data heterogeneity to emulate real-world scenario. For a dataset with C classes and M clients, we sample a class distribution vector $p_m = (p_{m,1}, \dots, p_{m,C})$ for client m from $\text{Dir}_C(\alpha)$, where $\alpha = 0.5$ controls the heterogeneity. We assign a proportion $p_{m,c}$ of class- c samples to client m . The key implications here are based on the heterogeneity parameter suggesting that setting $\alpha = 0$, would represent high heterogeneity as clients hold data from few classes, and $\alpha = \infty$ represents near i.i.d. distribution, which is also used in [24] for baseline comparison. Our $\alpha = 0.5$ balances realism as it ensures that the clients see distinct class distributions.

C. Implementation

This study uses PyTorch framework to implement POST and the comparative baselines. The experiments were conducted on GPU clusters with A100 and RTX3090 GPUs. We use ADAM optimizer with default values and a learning rate of $10e^{-3}$ for the datasets and the batch size was set to 32. We set the number of clients to ten, and for each round only five clients would participate chosen in a random manner. The communication rounds are set to 100. We set the adversarial

loss weight to 0.15. Hyperparameters were tuned through grid search: $\lambda_1(\text{sparsity}) \in [0.001, 0.1]$, $\lambda_2(\text{L2-mask stability}) \in [0.01, 0.1]$, $\lambda_3(\text{communication}) = 0.01$, $\lambda_4(\text{energy}) = 0.001$, and δ (adversarial weight) $\in [0.1, 0.2]$. The optimal values, i.e. $\lambda_1 = 0.01$, $\lambda_2 = 0.05$, and $\delta = 0.15$, were selected based on the minimized prediction loss on a 10% validation split of VERIWILD. The energy cost term (E) incorporated device-specific FLOP profiles (e.g., 0.5 W/GLOPS for edge device) [9], [24]. For the evaluation, we validate the performance of the proposed system using prediction accuracy, PSNR, SSIM, L2 norm distance, and attack accuracy, respectively.

D. Performance Analysis

We first evaluate the efficacy of the proposed POST on two datasets, comparing it with the existing FL, SL, and SFL approaches without any defense against inversion attack. It is assumed that if the original model is used, the E-IoT device can only implement two residual blocks without defense. However, the POST applies pruning, such that around two-third of the parameters are pruned, and therefore, the E-IoT devices can use four residual blocks. We evaluate the performance of the existing approaches and the proposed POST using SSIM drop, PSNR drop, prediction accuracy drop, and attack accuracy drop. The term “drop” in the performance analysis refers to the degradation of performance. The SSIM and PSNR drop represent the impact of inversion attack, such that higher SSIM and PSNR drop suggest better defense against inversion attacks. On the other hand, prediction accuracy drop represents the performance of the prediction system. A higher drop would indicate that the system is unable to predict the task accurately; therefore, the tradeoff is not suitable for practical implementation. The attack accuracy is considered to be successful if the reconstruction image can be categorized same as the target label. It represents the drop in the success rate of attack accuracy, suggesting that a higher value represents higher resilience to inversion attacks. We report the performance analysis in Table I. The proposed POST significantly reduces the impact of inversion attacks in terms of attack, SSIM, and PSNR accuracy, while maintaining

TABLE II
COMPARATIVE ANALYSIS ON RECONSTRUCTION
RESULTS GENERATED VIA INVERSION ATTACK

Method	Dataset	SSIM	PSNR	Prediction
NO Defense	COVID-19	0.54	21.36	92%
	VERIWILD-S	0.45	17.4	94%
	VERIWILD-L	0.45	17.4	94%
APP	COVID-19	0.46	18.4	92%
	VERIWILD-S	0.33	16.2	89.7%
	VERIWILD-L	0.34	16.1	79.2%
ZETA	COVID-19	0.48	18.8	91.3%
	VERIWILD-S	0.44	17.6	89.9%
	VERIWILD-L	0.43	17.2	79.7%
Drop Out Defense	COVID-19	0.46	18.7	93.7%
	VERIWILD-S	0.42	17.9	92.3%
	VERIWILD-L	0.41	17.8	87.4%
POST	COVID-19	0.33	14.5	94.2%
	VERIWILD-S	0.25	12.3	92.8%
	VERIWILD-L	0.22	11.7	88.7%

TABLE III
COMPONENTWISE ANALYSIS FOR THE POST METHOD
ON THE COVID-19 AND VERIWILD DATASETS

Modules	Dataset	SSIM drop	PSNR drop	Acc. drop
POST-R-Adv	COVID-19	8.9%	8.7%	7.5%
	VERIWILD	11.1%	10.9%	4.9%
POST+R-Adv	COVID-19	11.3%	10.8%	7.9%
	VERIWILD	13.5%	13.1%	5.3%
POST+Adv-R	COVID-19	12.6%	11.2%	7.1%
	VERIWILD	14.8%	13.3%	4.5%
POST+Adv+R	COVID-19	12.7%	12.3%	5.3%
	VERIWILD	14.9%	14.5%	2.7%

a reasonable prediction accuracy and effectively preserving user's personal information.

E. Comparative Analysis

We compare our approach with existing adversarial privacy preservation (APP) method [36], ZETA [20], and dropout defense method [14], respectively. We compare them with POST using PSNR, SSIM, and prediction accuracy. We repeated experiments several times for the existing methods to optimize the prediction accuracy. Therefore, the results reported for the PSNR and SSIM are when these methods achieved the best prediction accuracy as shown in Table II. The results show that the proposed POST method achieves least privacy risk while attaining better prediction performance in comparison to the existing approaches. The results show that the POST is not only suitable for real-world implications but also is secure. During the experiments, it was observed that the noise defense could achieve better privacy results by adding more noise; however, while adding the noise, the prediction performance decreases drastically. Similarly, the skip defense performs well if we skip more neurons; however, skipping neurons decrease the prediction performance. The APP performs better than the other defenses but the POST achieves the best tradeoff overall.

F. Ablation Study

As the proposed POST methods uses regularization and adversarial training in addition to the proposed pruning

TABLE IV
AVERAGE COMPUTATION LOAD IN TFLOPS AND AVERAGE
COMMUNICATION LOAD IN GB FOR 1 ROUND PER CLIENT

Method	TFLOPs	Bytes (GB)
FL	8.00	0.20
SL	1.14	5.78
SFL	1.14	5.78
SplitAvg	0.25	1.46
FedProx	8.18	0.21
POST	8.24	<u>3.45</u>

method, we conduct an ablation study to show that the components support positively to the performance of POST method. We report the results when pruning is performed without regularization and adversarial training, when the pruning is performed without regularization but with adversarial training, and when the pruning is performed with regularization but without adversarial training. The results are reported in Table III. For the first method, we only focus on the proposed structured pruning method without using regularization and adversarial training approach. The results show that even with only pruning, the POST method achieves better results than existing FL and SL paradigm in terms of privacy and shows on par performance in terms of prediction accuracy drop. The results also show that each of the component contributes positively to the privacy preservation. By integrating both the modules, we achieve the best results in relation to the componentwise and comparative results with the existing methods.

G. Communication and Computation Cost

Finally, we provide analysis concerning the communication and computation cost of the proposed POST and compare it with existing FL and SL paradigms. First, we determine the amount of data communication per round and then we compute the computation in terms of floating point operations per round at the E-IoT side. We report and compare the communication and computation cost for POST and other methods in Table IV. The communication in the proposed scenario refers to the uploading and downloading of models or transmitting the intermediate feature maps. As the model only transmits the intermediate features instead of model, the SL and SFL paradigm has an upper hand in the communication load. However, it should be noted that the SL and SFL only use two residual blocks due to their storage constraints at the E-IoT device, whereas the proposed POST uses 4 residual blocks. Despite that, the POST achieves comparatively better communication load capacity in comparison to existing approaches. Similarly, for the computation, SL and SFL cannot perform a lot of computation locally; however, the FL-based methods have a better efficiency in terms of computing. The proposed POST method prunes almost two-third of the parameters, which makes the computation very fast as observed in the results. We suggest that the proposed POST achieves the best tradeoff in terms of computational and communication load while preserving the privacy of the users' data.

H. Sensitivity Analysis

To test the proposed POST under varying pruning ratios, i.e., 40%–80%. The results for the sensitivity analysis are

TABLE V
SENSITIVITY ANALYSIS WITH RESPECT TO PRUNING
RATIOS ON THE VERIWILD DATASET

Pruning Ratio	SSIM Drop	PSNR Drop	Accuracy Drop	Attack Success	FLOPs Reduction
40%	0.102	10.5%	3.4%	15.2%	52%
50%	0.124	11.9%	4.9%	11.3%	58%
60%	0.147	15.0%	5.3%	5.7%	65%
70%	0.39	29.3%	8.9%	3.3%	72%
80%	0.41	30.1%	12.7%	3.1%	78%

reported in Table V. We evaluate for the varying pruning ratios using SSIM drop, PSNR drop, accuracy drop, attack success rate, and FLOPs reduction. The results reveal that the 60% pruning remains optimal, balancing attack resilience, and computational feasibility. It is apparent from the results that at 50% pruning, attack success drops to 11.3%, in comparison to 15.2%–40% pruning, with minimal accuracy degradation, i.e. 5.6%. The pruning ratio of 70% further reduces attack success to 3.3% but increases accuracy drop to 8.9% due to oversparsification. The 60% pruning also achieves the best balance, i.e., 65% FLOPs reduction along with 5.7% attack success, and ~5% accuracy drop. Beyond 70% pruning, accuracy degradation outweighs privacy gains, i.e., 12.7% drop at 80% pruning. It was also noted that higher pruning ratios amplify PSNR drop indicating severely distorted reconstructions.

V. CONCLUSION

In this article, we propose a distributed learning framework, named POST, to address the issues of inversion attacks for E-IoT devices. We propose a novel pruning strategy that could help reside more number of layers at the edge side while improving the communication and computational load. The constraints added in the optimization function helps improve the computations locally while reducing the communication, i.e. sparse intermediate feature maps, between the client and the cloud server. We also adopt the regularization and adversarial training strategy that improves the security and privacy of local edge users without compromising on the predictive performance. Our experiments are conducted on the real-world dataset, which shows that the proposed POST is applicable to real-world scenarios and achieves the best tradeoff in terms of computation and communication load.

Although we proposed a general framework to handle inversion and poisoning attack, in this work, we only focused on addressing the inversion attack. However, as a future work, we intend to extend this approach by defining the defense methods against poisoning attacks through the checking of pruned models, thus developing a trust metric to preserve the integrity of services offered to edge users.

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